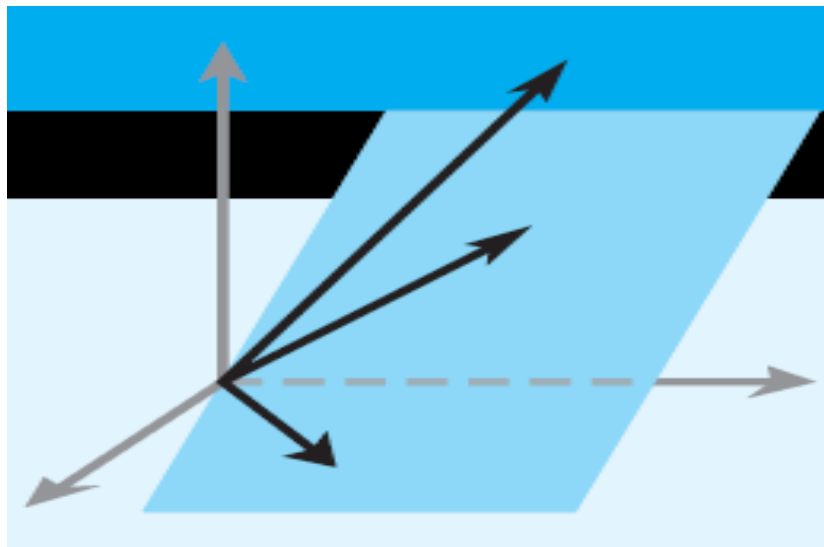


Elementary Linear Algebra

Chapter 7



Howard Anton
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Chapter 7

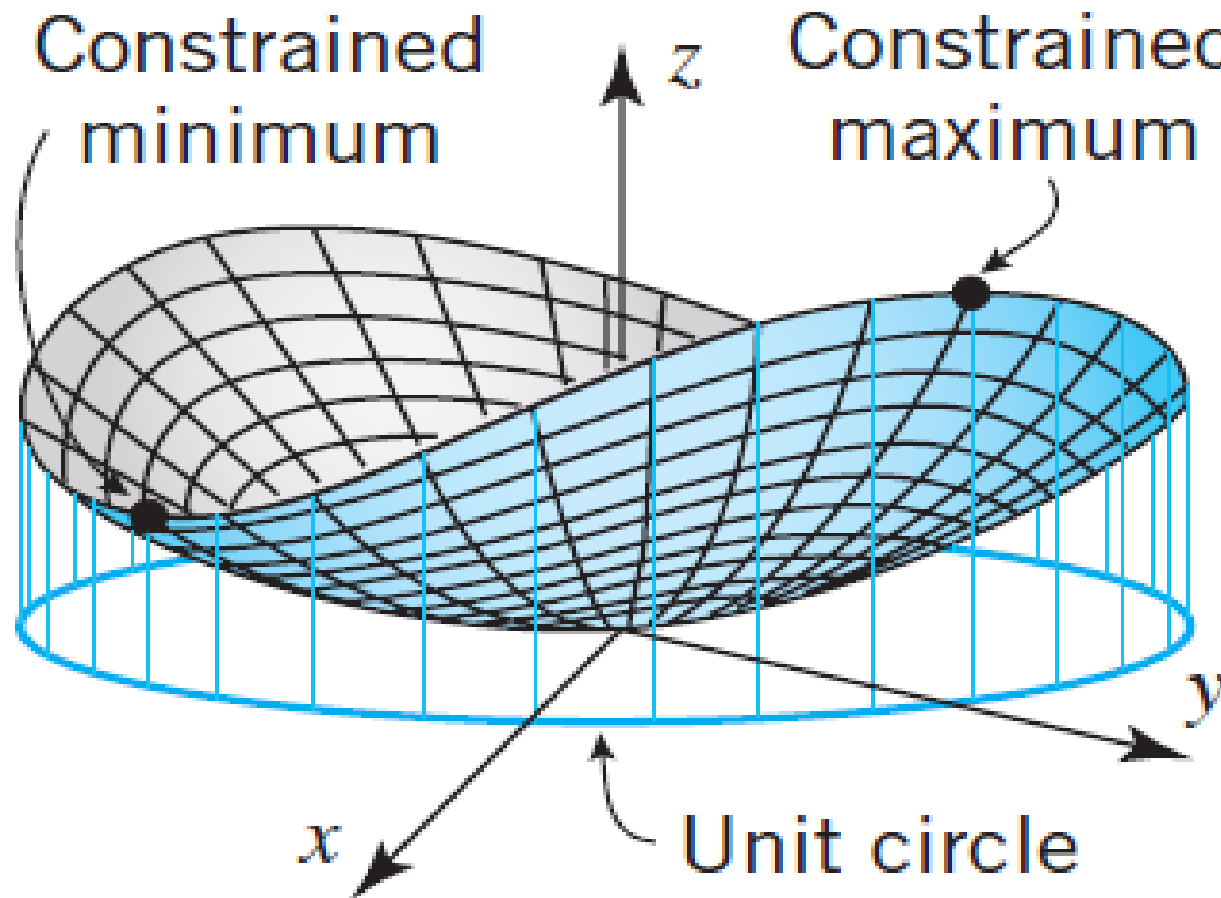
Diagonalization & Quadratic Forms

- **7.1 Orthogonal Matrices**
- **7.2 Orthogonal Diagonalization**
- **7.3 Quadratic Forms**
- **7.4 Optimization Using Quadratic Forms**
- **7.5 Hermitian, Unitary, and Normal Matrices**

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Section 7.4 Optimization Using Quadratic Forms

Question?



Section 7.4 Optimization Using Quadratic Forms

THEOREM 7.4.1 Constrained Extremum Theorem

Let A be a symmetric $n \times n$ matrix whose eigenvalues in order of decreasing size are $\lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_n$. Then:

- (a) the quadratic form $\mathbf{x}^T A \mathbf{x}$ attains a maximum value and a minimum value on the set of vectors for which $\|\mathbf{x}\| = 1$;*
- (b) the maximum value attained in part (a) occurs at a unit vector corresponding to the eigenvalue λ_1 ;*
- (c) the minimum value attained in part (a) occurs at a unit vector corresponding to the eigenvalue λ_n .*

Proof Page 418

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Theorem 4.7.1

Proof Page 418

A is symmetric matrix then A can be orthogonal diagonalizable

$$P^T A P = D$$

by Principal axes theorem

$$\begin{aligned} & \mathbf{x}^T A \mathbf{x} \\ &= \mathbf{y}^T D \mathbf{y} & P^T \mathbf{x} = \mathbf{y} \\ &= \lambda_1 y_1^2 + \lambda_2 y_2^2 + \cdots + \lambda_n y_n^2 \end{aligned}$$

$$\|\mathbf{x}\| = 1 \quad \longrightarrow \quad \|\mathbf{y}\| = \|P^T \mathbf{x}\| = \|\mathbf{x}\| = 1$$

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Theorem 4.7.1

$$\|y\|^2 = y_1^2 + y_2^2 + \cdots + y_n^2 = 1$$

$$\begin{aligned} x^T A x &= \lambda_1 y_1^2 + \lambda_2 y_2^2 + \cdots + \lambda_n y_n^2 \\ &\leq \lambda_1 y_1^2 + \lambda_1 y_2^2 + \cdots + \lambda_1 y_n^2 \end{aligned}$$

$$x^T A x \geq \lambda_n y_1^2 + \lambda_n y_2^2 + \cdots + \lambda_n y_n^2$$


$$\lambda_n \leq x^T A x \leq \lambda_1$$

Section 7.4 Optimization Using Quadratic Forms

► EXAMPLE 1 Finding Constrained Extrema

Find the maximum and minimum values of the quadratic form

$$z = 5x^2 + 5y^2 + 4xy$$

subject to the constraint $x^2 + y^2 = 1$.

Solution

The quadratic form can be expressed in matrix notation as

$$z = 5x^2 + 5y^2 + 4xy = \mathbf{x}^T A \mathbf{x} = \begin{bmatrix} x & y \end{bmatrix} \begin{bmatrix} 5 & 2 \\ 2 & 5 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$$

We leave it for you to show that the eigenvalues of A are $\lambda_1 = 7$ and $\lambda_2 = 3$ and that corresponding eigenvectors are

$$\lambda_1 = 7: \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \quad \lambda_2 = 3: \begin{bmatrix} -1 \\ 1 \end{bmatrix}$$

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Normalizing these eigenvectors yields

$$\lambda_1 = 7: \begin{bmatrix} \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{bmatrix}, \quad \lambda_2 = 3: \begin{bmatrix} -\frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{bmatrix} \quad (1)$$

Thus, the constrained extrema are

constrained maximum: $z = 7$ at $(x, y) = \left(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}\right)$

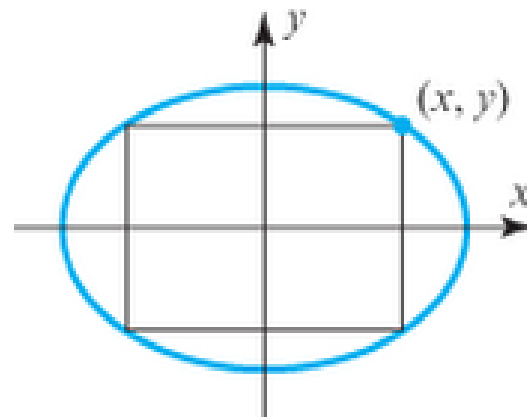
constrained minimum: $z = 3$ at $(x, y) = \left(-\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}\right)$



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► EXAMPLE 2 A Constrained Extremum Problem

A rectangle is to be inscribed in the ellipse $4x^2 + 9y^2 = 36$, as shown in Figure [7.4.2](#). Use eigenvalue methods to find nonnegative values of x and y that produce the inscribed rectangle with maximum area.



► **Figure 7.4.2**

A rectangle inscribed in the ellipse $4x^2 + 9y^2 = 36$.

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Solution

The area z of the inscribed rectangle is given by $z = 4xy$, so the problem is to maximize the quadratic form $z = 4xy$ subject to the constraint $4x^2 + 9y^2 = 36$. In this problem, the graph of the constraint equation is an ellipse rather than the unit circle as required in Theorem [7.4.1](#), but we can remedy this problem by rewriting the constraint as

$$\left(\frac{x}{3}\right)^2 + \left(\frac{y}{2}\right)^2 = 1$$

and defining new variables, x_1 and y_1 , by the equations

$$x = 3x_1 \quad \text{and} \quad y = 2y_1$$

This enables us to reformulate the problem as follows:

$$\text{maximize } z = 4xy = 24x_1y_1$$

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subject to the constraint

$$x_1^2 + y_1^2 = 1$$

To solve this problem, we will write the quadratic form $z = 24x_1y_1$ as

$$z = \mathbf{x}^T \mathbf{A} \mathbf{x} = \begin{bmatrix} x_1 & y_1 \end{bmatrix} \begin{bmatrix} 0 & 12 \\ 12 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ y_1 \end{bmatrix}$$

We now leave it for you to show that the largest eigenvalue of $\mathbf{A}$ is $\lambda = 12$ and that the only corresponding unit eigenvector with nonnegative entries is

$$\mathbf{x} = \begin{bmatrix} x_1 \\ y_1 \end{bmatrix} = \begin{bmatrix} \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{bmatrix}$$

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Thus, the maximum area is $z = 12$, and this occurs when

$$x = 3x_1 = \frac{3}{\sqrt{2}} \quad \text{and} \quad y = 2y_1 = \frac{2}{\sqrt{2}}$$



Section 7.4 Optimization Using Quadratic Forms

► EXAMPLE 3 Example 1 Revisited Using Level Curves

In Example 1 (and its following remark) we found the maximum and minimum values of the quadratic form

$$z = 5x^2 + 5y^2 + 4xy$$

subject to the constraint $x^2 + y^2 = 1$. We showed that the constrained maximum is $z = 7$, which is attained at the points

$$(x, y) = \left(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}} \right) \quad \text{and} \quad (x, y) = \left(-\frac{1}{\sqrt{2}}, -\frac{1}{\sqrt{2}} \right) \quad (3)$$

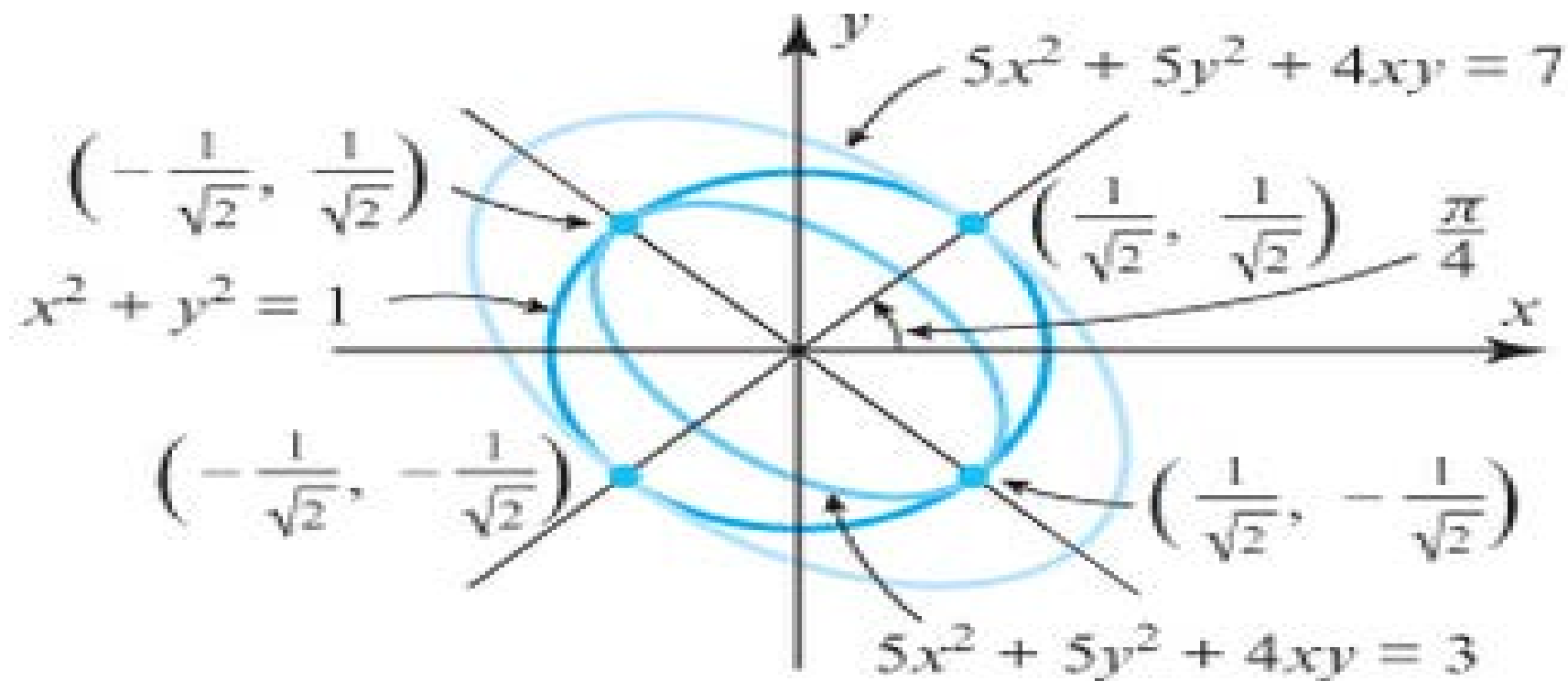
and that the constrained minimum is $z = 3$, which is attained at the points

$$(x, y) = \left(-\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}} \right) \quad \text{and} \quad (x, y) = \left(\frac{1}{\sqrt{2}}, -\frac{1}{\sqrt{2}} \right) \quad (4)$$

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Geometrically, this means that the level curve $5x^2 + 5y^2 + 4xy = 7$ should just touch the unit circle at the points in [3](#), and the level curve $5x^2 + 5y^2 + 4xy = 3$ should just touch it at the points in [4](#). All of this is consistent with Figure [7.4.5](#).

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► **Figure 7.4.5**

Section 7.4 Optimization

Using Quadratic Forms

THEOREM 7.4.2 Second Derivative Test

Suppose that (x_0, y_0) is a critical point of $f(x, y)$ and that f has continuous second-order partial derivatives in some circular region centered at (x_0, y_0) . Then:

(a) f has a relative minimum at (x_0, y_0) if

$$f_{xx}(x_0, y_0)f_{yy}(x_0, y_0) - f_{xy}^2(x_0, y_0) > 0 \quad \text{and} \quad f_{xx}(x_0, y_0) > 0$$

(b) f has a relative maximum at (x_0, y_0) if

$$f_{xx}(x_0, y_0)f_{yy}(x_0, y_0) - f_{xy}^2(x_0, y_0) > 0 \quad \text{and} \quad f_{xx}(x_0, y_0) < 0$$

(c) f has a saddle point at (x_0, y_0) if

$$f_{xx}(x_0, y_0)f_{yy}(x_0, y_0) - f_{xy}^2(x_0, y_0) < 0$$

(d) The test is inconclusive if

$$f_{xx}(x_0, y_0)f_{yy}(x_0, y_0) - f_{xy}^2(x_0, y_0) = 0$$

Section 7.4 Optimization Using Quadratic Forms

Our interest here is in showing how to formulate the above theorem using properties of symmetric matrices. Considering

$$H(x, y) = \begin{bmatrix} f_{xx}(x, y) & f_{xy}(x, y) \\ f_{xy}(x, y) & f_{yy}(x, y) \end{bmatrix}$$

which is called **the Hessian or Hessian matrix**

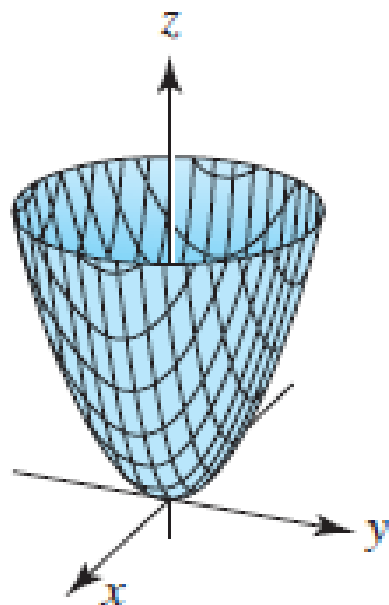
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THEOREM 7.4.3 Hessian Form of the Second Derivative Test

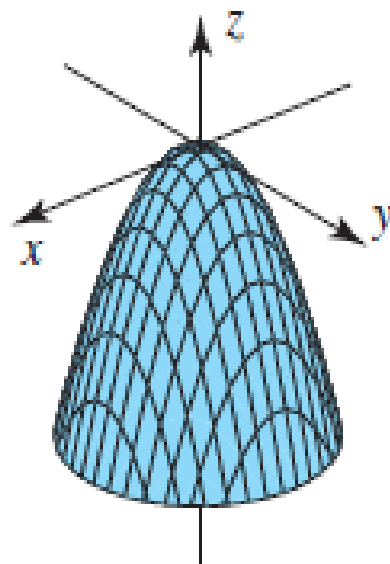
Suppose that (x_0, y_0) is a critical point of $f(x, y)$ and that f has continuous second-order partial derivatives in some circular region centered at (x_0, y_0) . If $H(x_0, y_0)$ is the Hessian of f at (x_0, y_0) , then:

- (a) f has a relative minimum at (x_0, y_0) if $H(x_0, y_0)$ is positive definite.*
- (b) f has a relative maximum at (x_0, y_0) if $H(x_0, y_0)$ is negative definite.*
- (c) f has a saddle point at (x_0, y_0) if $H(x_0, y_0)$ is indefinite.*
- (d) The test is inconclusive otherwise.*

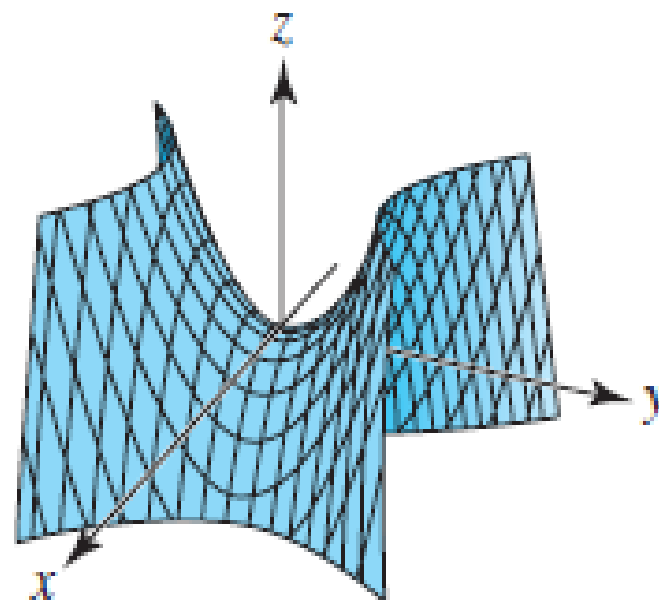
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Relative minimum at $(0, 0)$



Relative maximum at $(0, 0)$



Saddle point at $(0, 0)$

Section 7.4 Optimization

Using Quadratic Forms

► EXAMPLE 4 Using the Hessian to Classify Relative Extrema

Find the critical points of the function

$$f(x, y) = \frac{1}{3}x^3 + xy^2 - 8xy + 3$$

and use the eigenvalues of the Hessian matrix at those points to determine which of them, if any, are relative maxima, relative minima, or saddle points.

Solution

To find both the critical points and the Hessian matrix we will need to calculate the first and second partial derivatives of f . These derivatives are

$$\begin{aligned} f_x(x, y) &= x^2 + y^2 - 8y, & f_y(x, y) &= 2xy - 8x, & f_{xy}(x, y) &= 2y - 8 \\ f_{xx}(x, y) &= 2x, & f_{yy}(x, y) &= 2x \end{aligned}$$

Thus, the Hessian matrix is

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$$H(x, y) = \begin{bmatrix} f_{xx}(x, y) & f_{xy}(x, y) \\ f_{xy}(x, y) & f_{yy}(x, y) \end{bmatrix} = \begin{bmatrix} 2x & 2y-8 \\ 2y-8 & 2x \end{bmatrix}$$

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To find the critical points we set f_x and f_y equal to zero. This yields the equations

$$f_x(x, y) = x^2 + y^2 - 8y = 0 \quad \text{and} \quad f_y(x, y) = 2xy - 8x = 2x(y - 4) = 0$$

Solving the second equation yields $x = 0$ or $y = 4$. Substituting $x = 0$ in the first equation and solving for y yields $y = 0$ or $y = 8$; and substituting $y = 4$ into the first equation and solving for x yields $x = 4$ or $x = -4$. Thus, we have four critical points:

$$(0, 0), \quad (0, 8), \quad (4, 4), \quad (-4, 4)$$

Evaluating the Hessian matrix at these points yields

$$H(0, 0) = \begin{bmatrix} 0 & -8 \\ -8 & 0 \end{bmatrix}, \quad H(0, 8) = \begin{bmatrix} 0 & 8 \\ 8 & 0 \end{bmatrix}$$

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$$H(0,0) = \begin{bmatrix} 0 & -8 \\ -8 & 0 \end{bmatrix}, \quad H(0,8) = \begin{bmatrix} 0 & 8 \\ 8 & 0 \end{bmatrix}$$

$$H(4,4) = \begin{bmatrix} 8 & 0 \\ 0 & 8 \end{bmatrix}, \quad H(-4,4) = \begin{bmatrix} -8 & 0 \\ 0 & -8 \end{bmatrix}$$

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We leave it for you to find the eigenvalues of these matrices and deduce the following classifications of the stationary points:

Critical Point (x_0, y_0)	λ_1	λ_2	Classification
$(0, 0)$	8	-8	Saddle point
$(0, 8)$	8	-8	Saddle point
$(4, 4)$	8	8	Relative minimum
$(-4, 4)$	-8	-8	Relative maximum



Section 7.1 Orthogonal Matrices

Homework

Sec 7.4

True/False

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